COMP445 – Winter 2026
Data Communication & Computer Networks
Theory Assignment # 4
Due Date: Friday, April 10th, 2026 by 11:59PM

Warning: Redistribution or publication of this document or its text, by any means, is strictly prohibited. Additionally, publishing the solution publicly, at any point of time, will result in an immediate filing of academic misconduct.

1. Suppose that you are transmitting blocks of data over a link having a known bit error rate BER. What is the probability that a received block of size L has error(s) but goes undetected? What is this probability if a 1-bit parity is added? Explain your assumptions.

2. In an 802.11 wireless LAN using CSMA/CA, station S sends a data packet to station R (which is within range). Even if we assume that connectivity is symmetric (A can hear B if and only if B can hear A), and collisions are the only cause of errors, and RTS/CTS packets never collide with each other, it is still possible that R will fail to receive the data packet. Describe one way this might happen.

3. Let A and B be two stations attempting to transmit on an Ethernet. Each has a steady queue of frames ready to send; A's frames will be numbered Al, A2, and so on, and B's similarly. Let T = 51.2 μs be the exponential backoff base unit.
   Suppose A and B simultaneously attempt to send frame 1, collide, and happen to choose backoff times of 0 x T and 1 x T, respectively, meaning A wins the race and transmits Al while B waits. At the end of this transmission, B will attempt to retransmit B1 while A will attempt to transmit A2. These first attempts will collide, but now A backs off for either 0 x T or 1 x T, while B backs off for time equal to one of 0 x T, ..., 3 x T.
   a. Give the probability that A wins this second backoff race immediately after this first collision; that is, A's first choice of backoff time k x 51.2 is less than B's.
   b. Suppose A wins this second backoff race. A transmits A3, and when it is finished, A and B collide again as A tries to transmit A4 and B tries once more to transmit Bl. Give the probability that A wins this third backoff race immediately after the first collision.

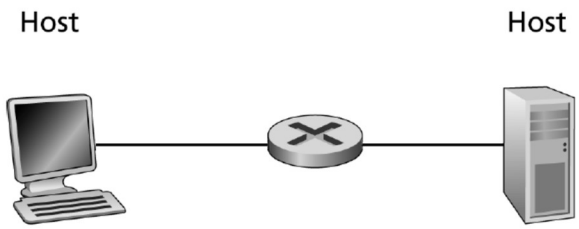
4. In the lecture, we provided an outline of the derivation of the efficiency of slotted ALOHA. In this problem we'll complete the derivation.
   a. Recall that when there are N active nodes, the efficiency of slotted ALOHA is $Np(1 - p)^{N-1}$. Find the value of p that maximizes this expression.

Redistribution or publication of this document or its text, by any means, is strictly prohibited.

b. Using the value of p found in (a), find the efficiency of slotted ALOHA by letting N approach infinity. Hint: $(1 - 1/N)^N$ approaches *1/e*, as N approaches infinity.

5. Consider the 4-bit generator G = 1001 is used in a Cycling Redundancy Check. Suppose that D has the value of 111010. What is the value of R?

6. Consider transmitting a packet from host A to host B via a router, as shown below:



Suppose that before sending the packet, all the ARP tables (in the two hosts and in the router) are empty. Let x denote the time to transmit a packet. Let y denote the delay from beginning the transmission of an ARP query until receiving and processing an ARP response. Ignore propagation delays. Assuming host A knows the IP address of host B, what is the total delay in moving the packet from Host A to Host B?

7. Consider an Ethernet LAN consisting of N nodes interconnected with a switch. Suppose the switch's forwarding table is initially empty. Suppose node A wants to TCP three-way handshake with node B, where both nodes are on the LAN. Assuming this is the only traffic on the network, and there are no packet errors or loss, how many frames will be transmitted in the process of establishing the TCP connection? Assume node A knows the IP address of node B, and ARP tables have all the necessary mappings.

Redistribution or publication of this document or its text, by any means, is strictly prohibited.

AI Use Disclosure Requirement

Now that you have completed your assignment, you must disclose and document any use of AI tools. This requirement applies to **everyone,** including students who did not rely on AI while solving the questions. The goal is to ensure transparency and fairness across all submissions.

To complete this component follow these instructions:

1. **Allowed Tools.** Use ChatGPT, Gemini, or Claude only through BassiliChat.com. Accessing these models elsewhere, or using any other AI model, is not permitted.
2. **If you used AI while solving the assignment**. For every question where AI helped you:
   - List all prompts, search commands, and interactions you used.
   - If you had an initial answer, compare it to the AI's answer, pointing out the key differences.
   - State which answer you believe is more accurate (yours or the AI's) and explain why.
3. **If you did not use AI while solving the assignment.** You must still work with AI for two questions of your choice. For each selected question:
   - Ask the AI to evaluate or corroborate your solution.
   - Explain whether the AI confirmed your answer or suggested changes.
   - Describe any additional insights the AI provided, even if your answer was already correct.
4. **Watch for AI Hallucinations**. AI tools occasionally generate incorrect or misleading content. If this occurs, identify the issue and explain why the AI's output is wrong.
5. **Restrictions on AI Prompts.** "Direct-answer" prompts are not allowed. These include:
   - Asking the AI to complete the entire assignment.
   - Requesting final solutions, reports, or code without your own reasoning.
   - Submitting ready-made answers with no evidence of your own work.

Using these types of prompts will result in **a grade of zero.** Your submission must demonstrate your own understanding and analysis, with AI acting only as a tool you reflect upon—not a substitute for your work.

Redistribution or publication of this document or its text, by any means, is strictly prohibited.